PERTANDINGAN WORLDSKILLS MALAYSIA KATEGORI BELIA (WSMB) TAHUN 2026

(IT NETWORK SYSTEM ADMINISTRATION)

MODUL B

WINDOWS ENVIRONMENT

PERINGKAT AKHIR

MASA : 6 JAM

NAMA: ____

No. KP: ____

JABATAN PEMBANGUNAN KEMAHIRAN

KEMENTERIAN SUMBER MANUSIA

Contents

Introduction to Test Project

The following is a list of sections or information that must be included in all Test Project proposals that are submitted to WorldSkills Malaysia Belia (WSMB).

- Contents including list of all documents, drawings and photographs that make up the Test Project
- Introduction/overview
- Short description of project and tasks
- Instructions to the Competitor
- Marking scheme (incl. assessment criteria)
- Other

Introduction

This Test Project proposal consists of the following documentation/files:

- ITNSA-FNAL-WSMB2026-Module-B-TestProject.pdf
- debian-12.5.0-amd64-DLBD-1.iso
- WindowsServer2019.iso
- ansible.cfg
- users.csv
- users.json

The competition has a fixed start and finish time. You must decide how to best divide
your time. Please **carefully read** the following instructions!

When the competition time ends, the machines in the manual state will not be changed. Make sure the PC is powered on. The machines involved in the automation except MGMT will be reverted to initial state for marking purposes.

Set all user password and credential with **Skills39** unless being specifically stated with password.

Description of project and tasks

You are the IT consultant responsible for WSMB 2026. You have to build and configure a windows network for the wsmb2026.my domain, create the users to this new domain and also implement features for external access to the network, policies and file services.

WSMB 2026 also requires some degree of scalability in their system, hence they opt in for an on-prem open source IaC automation solution as part of their environment. A Linux server with Ansible is installed in the topology that acts as the management server for the IaC part.

Default login for all VMs and Devices:
Username Linux: root / **itnsa**
Username Windows: Administrator

Password: **Skills39**

If you have difficulty or are unable to configure or work with a machine which is a “Core” server with
no GUI, you may re-install any machine as a server with the “desktop experience” at a small marking penalty
(and of course – the time it takes to carry out the install)

Part 1: HEADQUARTERS NETWORK

HQ-DC

Make sure the hostname and IP is set based on the topology. Make sure the device responds to ICMP packets.

Active Directory Domain Services (ADDS)

- Install the Active Directory Domain Services
  - Set this server as the primary domain controller and Global Catalogue for **wsmb2026.my**
- Create the following Organizational Units:
  - Managers
  - Competitors
  - Visitors
- Create the following global Active Directory groups under their respective OU.
  - WSMB_Managers
  - WSMB_Competitors
  - WSMB_Visitors
- Create a powershell script and save it in C:\create_user.ps1.
  - The script must create the following users according to the table below.

User Prefix	Numbers	Password	OU	Group
mgr	1-20	Skills@39	Managers	WSMB_Managers
cpt	1-20	Skills@39	Competitors	WSMB_Competitors
vis	1-20	Skills@39	Visitors	WSMB_Visitors

    - Example: cpt7, exp9, vis2, cpt18, exp11, vis20
  - The script should ensure that users created do not need to change password on next logon
  - Make sure every user is configured with their home folder share with the letter H:.
  - The script should ignore creating a user if it already exists.
  - The script should also take in argument of “-count”, when used with “-count X” where X is a number indicating the last number for user creation. If the value is not specified, then it defaults to 20.
    - Example: C:\create_user.ps1 -count 26 (Create user 1-26)
  - **Note:** If you are unable to create the powershell script the users can be created manually but you will not receive marks for the creation of the automated script.

Domain Name System (DNS)

- DNS service with name server of **wsmb2026.my** should be already installed alongside the ADDS
- Configure the primary **wsmb2026.my** name server
  - Create static A and PTR records for all servers and services of **wsmb2026.my** domain.
  - Create CNAME records required by services
- Configure root hint as “ns.msftncsi.com” and remove other root hints
- Configure split-brain DNS so that external DNS server is resolved to the external IP address.

Group Policy Object (GPO)

- Configure and apply the following group policies:
  - Ensure that all users will be greeted with a login banner and title with the following content:
    - Title: “Worldskills Malaysia Belia 2026”
    - Message: “Welcome to the competition of IT Network System Administration”
  - Disable the first sign-in animation for all domain users
  - Ensure that the folder \\wsmb2026.my\fileshare is mapped on all domain users with letter J:
  - Create a GPO called “visitors” to disable cmd, powershell and registry editor for domain users in the “WSMB_Visitors” group
  - Create a GPO called “managers” to automatically issue a certificate for the " WSMB_Managers" group members using the WSMB_Managers template.
  - Create a GPO policy called “work” which will automatically connect the work folder when "Competitors" group members logged on.

HQ-SRV

Make sure the hostname and IP is set based on the topology. Join this computer to wsmb2026.my domain. Make sure the device responds to ICMP packets.

Active Directory Certificate Services (ADCS)

- Install and configure the Subordinate Enterprise Certification Authority service
    - Use certificate issued from "ISP-CA".
    - Common Name: "SKILLS39-CA".
    - Enable extensions for CDP and AIA URL through HTTP on this server
        - CDP: http://cert.wsmb2026.my/CertEnroll/SKILLS39-CA.crl
        - AIA: http://cert.wsmb2026.my /CertEnroll/SKILLS39-CA.crt
    - Create the following templates:
        - “WSMB_Managers”
            - For the users in “WSMB_Managers” group.
        - “WSMB_Servers”
            - To provide PKI for the servers/services in wsmb2026.my domain

Internet Information Services (IIS)

- Install and configure the Internet Information Services (IIS)
    - Configure new site name “Website” for **https://www.wsmb2026.my** with the following content
        - “<h1><center>HQ-SRV Website</center></h1>”
- Traffic should be secured by public key generated by SKILLS39-CA.
- “rewrite_amd64_en-US.msi” should be provided to assist in this task
- Configure HTTP to HTTPS redirection for www.wsmb2026.my.
    - Do not send header that modify how browser interacts and handle future requests.
- All users accessing HTTPS webpage should not encounter any certificate errors in the browser.

Active Directory Federation Service (ADFS)

- Install and configure the Active Directory Federation Service (ADFS)
    - Configure the URL to be https://adfs.wsmb2026.my
    - Configure the display name to be “WSMB2026 Single Sign-On”
    - Enable idpInitiatedSignOn.aspx page.

HQ-FILE

Make sure the hostname and IP is set based on the topology. Join this computer to wsmb2026.my domain. Make sure the device responds to ICMP packets.

Dynamic Host Configuration Protocol (DHCP)

- Configure DHCP server for Internal Network and also Backend Network
  - Use the IP assignment range of **172.20.100.150 – 200 for Internal Network**
  - Set appropriate value of for scope options of DNS and Gateway

RAID

- Add 3 extra 1 GB drives to the VM
  - Format the attached disks with NTFS into a single RAID 5 array
  - Assign the drive to the letter G
  - Enable de-duplication on this volume.
    - Make sure the throughput optimization runs at 00:00 every day
    - Make sure background optimization is enabled

Work Folder

- Configure Work Folders. All domain users should be able to use this work folder.
  - Access URL: https://work.wsmb2026.my/
  - Local path: "G:\work\"

Home Folder

- Create file share for user's home drives.
  - Access UNC path: \\STORAGE.wsmb2026.my\homes
  - Local path: "G:\homes\"
- Limit home folders so that users cannot store more than 10 MB of data and cannot save bitmap (*.bmp)

File Sharing

- Create a writable file share by Domain Users in G:\public and share it as \\HQ-FILE\public
- Create a file share for local path G:\WSMB and share it as \\HQ-FILE\WSMB
  - Create two subfolders inside G:\WSMB\ and share and configure access control as follows:
    - Create a “Belia” folder.
      - Allow read-only access for users who is in "Competitors" OU
      - Allow full access to the users who belongs to “WSMB_Managers” group
    - Create a “Pertandingan” folder.
      - Allow read-only access to “Visitors” OU
      - Allow full access to the users who belongs to “WSMB_Managers” group
  - The folders should be hidden for all users who have insufficient permission to access the folder.
  - All folders should be accessible for Administrators.

- Install and configure DFS so that the “WSMB” and “public” shares are accessible by accessing \\wsmb2026.my\fileshare

HQ-EDGE

Make sure the hostname and IP is set based on the topology. Join this computer to wsmb2026.my domain. Make sure the device responds to ICMP packets.

Routing and Remote Access (RRAS)

- Enable Routing.
- NAT necessary services for public access
- Internal servers should be able to access the INET server.
- Users and computers on the Internet should be able to establish VPN connection to this server.
  - Authorize VPN access through the NPS.
  - IP address pool for remote access clients: 192.168.200.55 - 192.168.200.110

IKEv2 VPN

- Setup a IKEv2 IPSec Site-to-Site VPN tunnel between HEADQUARTERS and BRANCH.
  - Set the interface name as "VPN"
  - The authentication should be done using certificate generated by SKILLS39-CA.
  - Make the connection as persistent.

Web Application Proxy

- Install and configure Web Application Proxy (WAP)
  - Clients from another network should be able to access https://www.wsmb2026.my after passing the ADFS web authentication.

CLIENT

- Make sure the hostname and IP is set based on the topology.
- Make sure the device responds to ICMP packets.
- Make sure this server is joined to the domain of wsmb2026.my
- Test access to websites
- Test GPOs, home and shared folders
- Make sure PKI is trusted

Part 2: BRANCH NETWORK

BR-DC

Make sure the hostname and IP is set based on the topology. Make sure the device responds to ICMP packets.

Active Directory Domain Services (ADDS)

- Install the Active Directory Domain Services
  - Set this server as the child domain of **branch.wsmb2026.my**
- Create the following Organizational Units:
  - Branch
- Create the following global Active Directory groups under the same OU.
  - Branch
- Create the following users according to table

User Prefix	Numbers	Password	OU	Group
branch	1-2	Skills39	Branch	Branch

- Make sure computers joined in parent domain of wsmb2026.my are able to login to this user.

Domain Name System (DNS)

- DNS service with name server of **branch.wsmb2026.my** should be already installed alongside the ADDS
- Configure the primary **branch.wsmb2026.my** name server
  - Create static A and PTR records for all servers and services of **branch.wsmb2026.my** domain.
- Configure root hint as “ns.msftncsi.com” and remove other root hints

Dynamic Host Configuration Protocol (DHCP)

- Configure DHCP server for Internal Network and also Backend Network
  - Set appropriate value of for scope options below
    - Default gateway : 172.30.1.1
    - DNS server : 172.30.1.100
    - Lease range : 172.30.1.161 – 172.30.1.250
    - Scope name : Branch Client
    - Exclude range : 172.30.1.161 – 172.30.1.170
    - Duration : 5 days, 11 hours, 59 minutes

BR-EDGE

Make sure the hostname and IP is set based on the topology. Make sure the device responds to ICMP packets. Make sure this server is joined to the domain.

Routing and Remote Access (RRAS)

- Enable routing
- Internal servers should be able to access the INET server.

IKEv2 VPN

- Setup a IKEv2 IPSec Site-to-Site VPN tunnel between HEADQUARTERS and BRANCH.
  - Set the interface name as “VPN”
  - The authentication should be done using certificate generated by SKILLS39-CA

ANSIBLE-SRV

You have decided to use Ansible to manage these project shares. For convenience reason, you are using a Debian machine called ANSIBLE-SRV to run the Ansible playbooks. VS Code and Zeal Docs with Ansible offline docs are ready to use.

Your colleague wrote a script, which exported the file shares and the corresponding AD groups as a YAML file (/etc/ansible/group_vars/windows). This file might be changed during marking to ensure that the script properly obtain values from this file.

The variable "file_shares" holds a list of file shares and their corresponding AD groups for the NTFS permission. The attribute "name" contains the Windows file share name, the attributes "read" & "write" hold the AD group for the corresponding NTFS permission.

This information is available as Ansible group variables. You are required to prepare the Ansible environment and the Ansible inventory that are relevant for Ansible to connect to Windows Servers.

- Configure a secure channel that is used by ANSIBLE-SRV to remotely manage BR-DC.
  - The channel must use HTTPS as the protocol and listen on port 5986.
  - When the secure channel has been configured, remove other non-secure channels
- Initialize a git repository in /opt/ansible/
  - Make sure to set the global username for git as your full name.
  - Use any of your preferred name with domain of @wsmb2026.my for the global email address.
- Create an Ansible playbook called /opt/ansible/manage-shares.yml on ANSIBLE-SRV to manage Windows shares on BR-DC
  - Create the domain AD groups used to control the access to the project shares
  - Create a folder C:\project_share and make it full access file shares on the corresponding server.
  - Create the corresponding sub-folders in the folder created above based on the defined values in the group variables.
  - Configure NTFS permissions on the subfolders based on the defined values in the group variables.
    - The key "read" means ReadAndExecute and the key "write" means Modify
  - Make sure that the Ansible playbook can run without entering any credentials. Marking process will execute the playbook within the /opt/ansible/ folder as follows:
    - ansible-playbook manage-shares.yaml
- Create a git commit with the message "final commit" with all your change

/etc/ansible/group_vars/windows

```
file_shares:
  share01:
    name: "wsmb_share01"
    read: "wsmbone"
    write: "wsmbtwo"
  share02:
    name: "wsmb_share02"
    read: "wsmbtwo"
    write: "wsmbthree"
file_groups:
  wsmbone:
  wsmbtwo:
  wsmbthree:
```

REMOTE

- Make sure the hostname and IP is set based on the topology. This PC will be used to also test INET.
- Make sure the device responds to ICMP packets.
- Make sure this server is joined to the domain of branch.wsmb2026.my.
- Make sure this computer is able to login to users in branch.wsmb2026.my domain.

Part 3: INET NETWORK

INET

Make sure the hostname and IP is set based on the topology. Make sure the device responds to ICMP packets.

Network Connectivity Status Indicator (NCSI)

- Configure so that clients should indicate that the network connection as “Internet”

Domain Name System

- Enable Routing Create zones and records for NCSI.
  - Add an A record "cs.msftncsi.com"
  - Add an A record "ns.msftncsi.com"
  - SOA and NS record of the "msftncsi.com" should be "ns.msftncsi.com".
- Create a root zone(.) to simulate the root DNS server.
  - Create appropriate delegations to resolve DNS records.

Dynamic Host Configuration Protocol (DHCP)

- Configure DHCP server for External Network
  - Use the IP assignment range of **1.9.250.101 - 105**
  - Set appropriate value of for scope options of DNS.
  - Gateway is set to the INET server itself

Active Directory Certificate Services (ADCS)

- Install and configure the Certification Authority service
  - Configure Standalone Root Certification Authority
  - Common Name: "ISP-CA".
  - Enable extensions for CDP and AIA URL through HTTP on this server
    - CDP: http://cs.msftncsi.com/CertEnroll/ISP-CA.crl
    - AIA: http:// cs.msftncsi.com /CertEnroll/ISP-CA.crt

Network Address Table

Device	Operating System	Address
INET	Windows Server 2022 GUI	1.9.250.100/28 202.188.1.100/24
HQ-DC	Windows Server 2022GUI	172.20.100.100/24
HQ-FILE	Windows Server 2022CORE	172.20.100.10/24
HQ-SRV	Windows Server 2022GUI	172.20.100.20/24
BR-DC	Windows Server 2022GUI	172.30.1.100/24
HQ-EDGE	Windows Server 2022GUI	172.20.100.1/24 [HQ] 1.9.250.97/28 [INET]
BR-EDGE	Windows Server 2022GUI	172.30.1.1/24 [BRANCH] 202.188.1.10/24 [INET]
ANSIBLE-SRV	Linux Debian 12.11 CLI	172.30.1.XXX/24 (DHCP)
CLIENT	Windows 11	DHCP Assignment
REMOTE	Windows 11	[INET-HQ]DHCP Assignment

Network Topology